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Review

Effect of carbonitrile and hexyloxy substituents on alternated copolymer of polythiophene–Performances in photovoltaic cells

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A novel series of regioregular bithiophene disubstituted with a carbonitrile group as electron acceptor and a hexyl or hexyloxy chain as electron donor, have been synthesized. The position and the regioregularity of substituents on bithiophene have been controlled via a head-to-head (HH) or head-to-tail (HT) coupling reaction to give four monomers: the HH and the HT carbonitrilehexylbithiophene monomers and the HH and the HT carbonitrilehexyloxybithiophene monomers. Each of them lead, by chemical polymerization, to four polymers constituted by a polythiophene backbone containing alternating electron-donating and electron-withdrawing units directly connected to the polythiophene. Optical and electrochemical characterizations reveal that all these polymers have a low band gap and absorb at longer wavelengths in comparison with a poly(3-hexylthiophene) (P3HT) film. The planarity of the polythiophene backbone was increased by introduction of sulphur–oxygen (S–O) interactions between hexyloxy chains and thiophene units. In addition, the introduction of carbonitrile groups allows to stabilize the LUMO and the HOMO energy levels of polymers at around -3.10 eV for the LUMO and $-5.0/-5.3$ eV for the HOMO, which were lower than the ones of P3HT (-2.19 eV for the LUMO and ~ -4.9 eV for the HOMO). Bulk heterojunction photovoltaic devices were fabricated using blends of these polymers with PCBM ([6,6]-phenyl-C61-butyric acid methyl ester). Power conversion efficiency of 1.07% was obtained under solar illumination AM 1.5 (100 mW cm^{-2}) from a device containing the polymer synthesized from HT 3-carbonitrile-4′-hexyloxyl-2,2′-bithiophene monomers and PCBM as the active layer.

Contents

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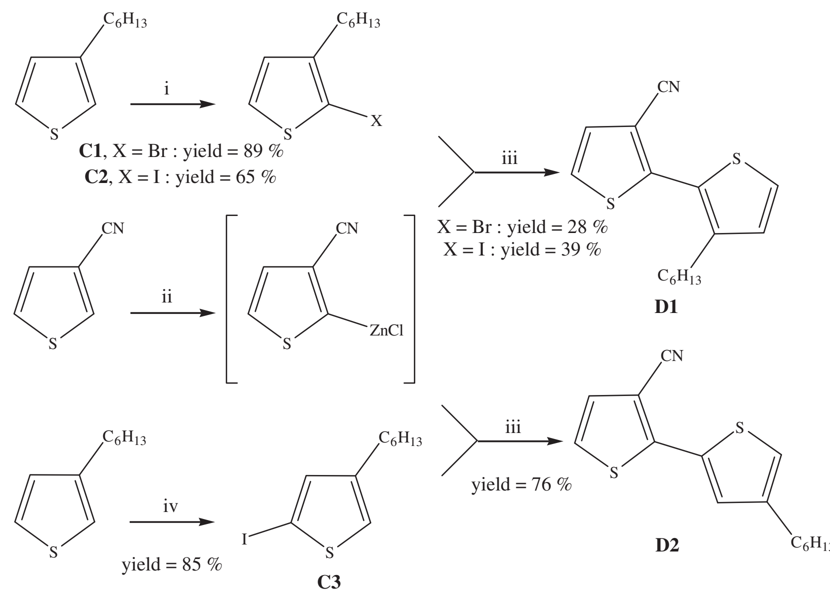
1. Introduction

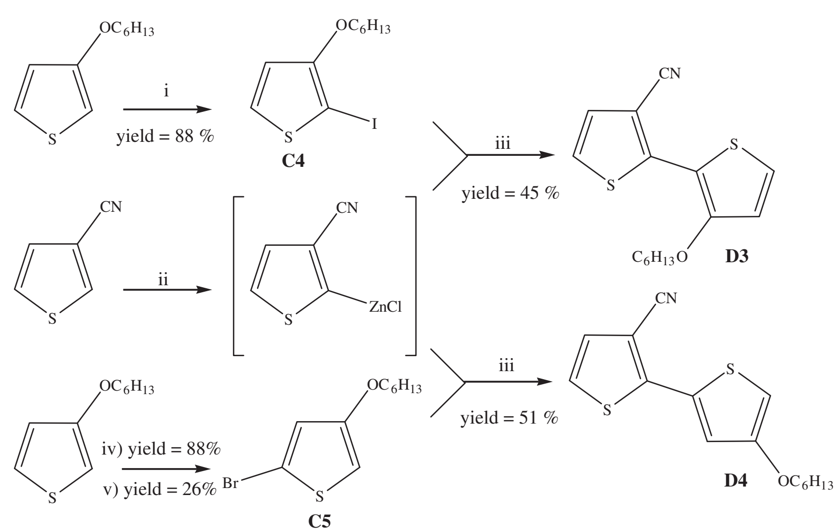
Conjugated polymers in their semi-conducting forms, are widely used within the organic electronic field for electronic applications like organic field-effect transistors [1,2] or optoelectronic applications like OLEDs or organic photovoltaics [3,4] (OPVs). At present time, the OPVs based on the bulk heterojunction concept allows to reach efficiencies up to 5% [5–7]. For this type of organic solar cells, the active layer is a blend of a conjugated polymer used as electron-donor material and a high-electron-affinity molecule like C_{60}. The cell with power conversion efficiency of 5% have been elaborated using a blend of regioregular poly(3-hexylthiophene) (P3HT) and [6,6]-phenyl-C61-butyric acid methyl ester (PCBM). The best photovoltaic properties obtained with P3HT should be attributed to major aspects like good absorption in the visible region and high charge mobility. Further improvements on the polymer solar cells require polymers that absorb more broadly and at longer wavelengths to match the terrestrial solar radiation spectrum. In order to make the polymers absorb more photons, it is necessary to minimize the band gap of the polymers to increase the photon absorption i.e. to synthesize low band gap polymers [3]. Besides band gap, several other characteristics of the conjugated polymers including HOMO/LUMO levels and charge carrier mobility need to be optimized simultaneously in order to achieve high photovoltaic performances. Indeed the difference between the HOMO level of the electron-donating polymer and the LUMO level of the PCBM should be maximised in order to increase the open-circuit voltage (V_{oc}) delivered by the photovoltaic devices [8]. In addition, the LUMO level of the polymer should be positioned above the LUMO of the PCBM (-4.3 eV) to an amount estimated to around 0.3 eV to ensure efficient electron transfer. To enhance the OPV performances, efficient charge transport within the interpenetrating networks is also required since the charge mobility could be a bottleneck in the performance of polymer solar cells. For the polymer-PCBM bulk heterojunction structure, a high charge mobility of conjugated polymers is essential for charge extraction in order to avoid a highly unbalanced charge transport in the blend that could be the result of a charge carrier mobility difference in the PCBM and polymer network.

Most of the photovoltaic polymers described in the literature are thiophene-based polymers either as a homopolymer or copolymer [9–12]. For example the alternation between thiophene as electron donor and benzothiadiazole as electron acceptor yields an electron donor–acceptor alternating copolymer having a low band gap and then can more efficiently absorb the red and near-infrared position of the solar emission spectrum [13]. For polythiophene, the optoelectronic features could be tuned thank to the incorporation of electron-withdrawing groups such as CN, NO_2, quinoxaline, and thiadiazole on the polythiophene chains. Furthermore the planarity of polythiophene structure has an influence on these properties. Indeed a planar structure generally exhibit a lower $\pi \rightarrow \pi^*$ transition. Planarity along the aromatic backbone results in a low band gap due to a high degree of delocalization of the π electrons [14]. In addition, a longer conjugation length in the polymer backbone is of great importance for charge carrier mobility and transport along polythiophene chains. It is known for the polythiophene that the introduction of alkoxy side chains on thiophene units allows to increase the coplanarity of the thienylene moieties along the polymer chain [15,16].

In this work, we present a family of four conjugated polymers synthesized by oxidative polymerization with $FeCl_3$ from unsymmetrical disubstituted bithiophene monomers. Our aim is the synthesis of conjugated polythiophenes in which the thiophene units are alternatively substituted with donor and acceptor substituents using hexyl or hexyloxy groups as electron donor and carbonitrile groups (CN) as electron acceptor. The substitution of polythiophene with both electron-donating and electron-withdrawing groups allows tailoring the energy levels of the polythiophene. Hence the band gap can be tuned in order to increase the harvesting of the sunlight by an expansion of absorption wavelength of the polymer in the visible region. We have chosen the CN group because it is well-known that its presence leads to a reduced band gap of the conjugated polymer [17–21] and we believe that it is small enough not to generate any distortions in the polythiophene backbone in order to keep its optimal planarity. Indeed in our approach, we have tried to combine the electronic effects introduced at the monomer level with geometrical effects on a large scale in order to obtain a polymer having both a low band gap and a coplanar architecture. In this perspective, we have synthesized the monomer 3-carbonitrile-4′-hexyl-2,2′-bithiophene (noted **D2** in Scheme 1) by a head-to-tail (HT) chemical coupling between the 3-carbonitrilethiophene and 4-hexylthiophene. The chemical polymerization of **D2** gives a polymer (noted $\mathbf{P_{D2}}$) for which the twisting between adjacent repeating units is minimized furthering the planarity of the polymer chains. In order to increase the planarity, we have also synthesized the polymer $\mathbf{P_{D4}}$ (Scheme 2) from 3-carbonitrile-4′-hexyloxyl-2,2′-bithiophene monomer (noted **D4** in Scheme 2). Thanks to sulphur–oxygen (S–O) interactions, the planarity of the polymer backbone should be improved since these interactions hold two adjacent thiophene moieties in a more or less fixed planar conformation. We describe herein the synthesis, characterization and optical and electrochemical properties of $\mathbf{P_{D2}}$ and $\mathbf{P_{D4}}$ polymers, which are compared with those of polymers (noted $\mathbf{P_{D1}}$ and $\mathbf{P_{D3}}$)



Scheme 1. Scheme of synthesis of 3-carbonitrile-3′-hexyl-2,2′-bithiophene **D1** and 3-carbonitrile-4′-hexyl-2,2′-bithiophene **D2**. (i) NXS, X=Br: DMF/MeOH, −50 to 25 °C, X=I: $CHCl_3/CH_3CO_2H$ (1:1), 0–25 °C; (ii) LDA, Et_2O, −78 to 25 °C, 1 h, $ZnCl_2$, THF, −78 to −50 °C, 2 h; (iii) $Pd(PPh_3)_3$, −50 to 25 °C, 2 h, reflux, 12 h; and (iv) BuLi, I_2, THF, −78 °C, 1 h.



Scheme 2. Scheme of synthesis of 3-carbonitrile-3′-hexyloxy-2,2′-bithiophene **D3** and 3-carbonitrile-4′-hexyloxy-2,2′-bithiophene **D4**. (i) NIS, $CHCl_3/CH_3CO_2H$ (1:1), 0–25 °C, 12 h; (ii) LDA, Et_2O, −78 to 25 °C, 1 h, $ZnCl_2$, THF, −78 to −50 °C, 2 h; (iii) $Pd(PPh_3)_3$, −50 to 25 °C, 2 h, reflux, 12 h; (iv) NBS, THF, −78 °C, 2 h; and (v) BuLi, Et_2O, −78 °C, 30 min., H_2O.

synthesized from bithiophene monomers elaborated with a head-to-head coupling (noted **D1** and **D3** in Schemes 1 and 2). Furthermore, the photovoltaic properties of all these polymers were investigated.

2. Experimental

2.1. Materials and instruments

All reagents were purchased from Aldrich. Tetrahydrofuran (THF) and propylene carbonate (PC) were dried and distilled prior use (THF from sodium benzophenone and PC from $CaCl_2$). Other reagents were used as received. ^1H and ^{13}C NMR spectra were recorded on a Bruker AC 200 MHz spectrometer and a VARIAN 400 MHz for polymers. Elemental analyses were carried out by the Analytical Service of CNRS of Vernaison (France). Molecular weights and polydispersities of polymers were determined by size exclusion chromatography (SEC) analysis on an 1100 HP Chemstation equipped with a 300 mm × 7.5 mm PLgel Mixed-D 5 μm10^{-4} Å column (Polymer Laboratories), a diode-array UV–vis detector and a refractive index detector. The column temperature and the flow rate were fixed to 40 °C and 1 mL min^{-1}. The calibration curve was constructed using ten polystyrene (PS) isodispersed standards (S-M2-10* kit from polymer Laboratories). Runs of 20 μL injection of dilute solutions of P3HT (2.5 mg g^{-1}) in HPLC-grade tetrahydrofuran were typically analysed for each sample with UV–vis detections at 420 nm. Other analyses are performed with a miniDAWN TREOS (Cell type: K5, and calibration constant: 4.3500 e−5 1/(V cm)) with a fixed wavelength at 658 nm with refractive index detector. The samples are prepared in THF (refractive index of 1.402) with a flow rate at 0.5 mL mn^{-1}. The mass molar are obtained first with a Dn/Dc of polythiophene of 0.24. They are also calculated in polystyrene equivalent with PS standard calibration.

Absorption spectra were recorded with a UVIKON XS spectrophotometer from Bio-TEK instruments. Fluorescence spectra were measured using a F-4500 spectrophotometer from Hitachi instruments. All the cyclic voltammetry experiments were carried out in argon atmosphere in a glove box, using three-electrode cell with a gold working electrode ($S=7\times10^{-2}$ cm^2), a platinum wire counter electrode and Ag/Ag^+ (0.01 M) reference electrode. Tetrabutylammonium tetrafluoroborate ($TBABF_4$) in an anhydrous propylene carbonate solution (0.1 mol L^{-1}) was used as the supporting electrolyte. Polymers to be measured were coated by drop casting on the gold working electrode from a chloroform solution of polymer (5 mg mL^{-1}). The electrochemical instrumentation constituted of an EG&G Princeton Applied Research potentiostat–galvanostat model 273A controlled by a computer with an EG&G software, which also collected the results.

Polymer photovoltaic cells were prepared in a glove box, on glass/(indium tin oxide) ITO (PGO, 20 Ω sq^{-1}). The substrates were cleaned in an ultrasonic bath with water and acetone. Then they were exposed to UV-ozone for 30 min. Poly(3,4-ethylenedioxythiophene) (PEDOT):poly(styrenesulfonate) (PSS) (Baytron PH) was deposited by spin coating and dried for 15 min at 150 °C. The thickness of PEDOT:PSS is about 40 nm. In a glove box, a freshly prepared mixture of polymer (CH_2Cl_2 fraction):PCBM solubilized in chlorobenzene was then deposited by spin coating on the ITO/PEDOT:PSS electrode to form the photosensitive layer. To finish, the LiF/Al electrode was deposited by evaporation onto the photosensitive layer. The active surface was 0.28 cm^2. All the measurements were performed in a glove box, under white light simulated illumination AM 1.5, 100 mW cm^{-2}. Current–voltage characteristics and power conversion efficiencies were measured via a computer controlled Keithley® SMU 2400 unit.

2.2. Monomer syntheses

The synthetic route to monomers is shown in Schemes 1 and 2. The detailed synthetic procedures are as follows:

2.2.1. 3-hexyloxythiophene **M1**

30 g of hexanol are added on 3.4 g of sodium (0.15 mol) under argon at ambient temperature and then heated to 110 °C during 5 h. The excess of hexanol was removed by distillation under vacuum. 0.7 equivalents of 3-bromothiophene (0.105 mol, 17.1 g) were added to the mixture with 10 mL of anhydrous DMF. The reaction mixture was heated to 110 °C for 5 h and 3 g of CuI was added by small portion all over the reaction time. The mixture was extracted with 3×50 mL of diethyl ether. The organic layers were washed with NH_4Cl, water and brine and dried over Na_2SO_4. After filtration and solvent evaporation, the product was purified by distillation. 22 g of yellowish oil were obtained (yield 86%).

^{1}H NMR ($CDCl_3$): 7.16 (q, 1H, J=5.1 Hz, J=3.1 Hz); 6.75 (q, 1H, J=5.1 Hz, J=1.6 Hz); 6.22 (q, 1H, J=3.2 Hz, J=1.6 Hz); 3.93 (t, 2H, J=6.3 Hz); 1.76 (m, 2H); 1.35 (m, 6H) and 0.90 (t, 3H, J=5.1 Hz). ^{13}C NMR ($CDCl_3$): 58.0; 124.4; 119.5; 96.9; 70.2; 31.6; 29.2; 25.7; 22.6 and 14.0.

2.2.2. 2-bromo-3-hexylthiophene **C1**

4.77 g (28.3 mmol) of 3-hexylthiophene was diluted in 10 mL of anhydrous DMF under argon at −50 °C. 1.2 equivalents (6.04 g, 34.0 mmol) of N-bromosuccinimide solubilised in 20 mL of DMF were added dropwise over a period of time of 30 min to the solution kept at −50 °C. The reaction mixture is allowed to warm up to room temperature in 2 h. The mixture was hydrolyzed and then extracted with hexane. The organic layers were washed successively with a saturated solution of $Na_2S_2O_3$, water and brine. They were dried over Na_2SO_4. The crude product was purified by distillation. 6.27 g (25.37 mmol) of colorless oil were obtained (yield 89.5%).

^{1}H NMR ($CDCl_3$): 7.16 (d, 1H, J=5.7 Hz); 6.78 (d, 1H, J=5.4 Hz); 2.55 (t, 2H, J=7.3 Hz); 1.55 (m, 2H); 1.29 (m, 6H); and 0.87 (t, 3H, J=6.7 Hz). ^{13}C NMR ($CDCl_3$): 141.9; 128.2; 125.1; 108.7; 31.6; 29.7; 29.4; 28.9; 22.6; and 14.1.

2.2.3. 3-hexyl-2-iodothiophene **C2**

2.00 g (11.9 mmol) of 3-hexylthiophene was diluted in 10 mL of anhydrous chloroform under argon at 0 °C with an ice bath. 1.0 equivalent (2.68 g, 11.9 mmol) of N-iodosuccinimide was solubilised in 60 mL of chloroform/acetic acid (1:1) and added dropwise to the solution kept at 0 °C. The reaction mixture was allowed to warm up to room temperature during the night (red coloration of the solution). The mixture was hydrolyzed and then extracted with hexane. The organic layers were washed with saturated solution of $Na_2S_2O_3$ and with water and brine. They were dried over Na_2SO_4. The crude product was purified by column chromatography on silica gel using hexane as eluent. A more efficient purification was performed with reverse phase HPLC (eluent: acetonitrile). 2.27 g (7.7 mmol) of colorless oil were obtained (yield 65%).

^{1}H NMR ($CDCl_3$): 7.36 (d, 1H, J=5.4 Hz); 6.73 (d, 1H, J=5.4 Hz); 2.53 (t, 2H, J=7.6 Hz); 1.53 (m, 2H); 1.29 (m, 6H) and 0.87 (t, 3H, J=6.4 Hz). ^{13}C NMR ($CDCl_3$): 147.14; 130.26; 127.93; 73.97; 32.08; 31.62; 30.00; 28.88; 22.60 and 14.10.

2.2.4. 4-hexyl-2-iodothiophene **C3**

752 mg (1 eq., 4.5 mmol) of 3-hexylthiophene was diluted in 7.5 mL of THF. The mixture was cooled down to −78 °C. 2.29 mL (1 eq., 4.5 mmol) of 1.95 mol L^{-1} nBuLi solution in hexane was added dropwise. The reaction mixture was allowed to warm up to room temperature for 1 h. 1.25 g of I_2 (1.1 eq., 4.92 mmol) were solubilized in 4 mL of THF and added to the mixture cooled down to −78 °C. The solution was kept at −78 °C during 1 h before being hydrolyzed with saturated sodium thiosulfate solution. The mixture was then allowed to warm up to room temperature. The mixture was extracted with 50 mL of hexane and 50 mL of saturated sodium thiosulfate solution. The organics were washed with saturated solution of $Na_2S_2O_3$ and with water. They were dried over Na_2SO_4, and the solvent was evaporated. The crude product was purified by column chromatography on silica gel using hexane as eluent. 1.12 g of yellowish oil was obtained (yield 85%).

^{1}H NMR ($CDCl_3$): 7.04 (d, 1H, J=1.6 Hz); 6.91 (d, 1H, J=1.6 Hz); 2.55 (t, 2H, J=7.6 Hz); 1.53 (m, 2H); 1.27 (m, 6H) and 0.86 (t, 3H, J=6.4 Hz). ^{13}C NMR ($CDCl_3$): 145.7; 138.4; 126.4; 73.1; 32.1; 30.7; 30.6; 29.3; 23.0 and 14.5.

2.2.5. 3-hexyloxy-2-iodothiophene **C4**

5 g (27.2 mmol) of 3-hexyloxythiophene was diluted in 30 mL of anhydrous chloroform under argon at 0 °C with an ice bath. 1.0 equivalent (6.12 g, 27.2 mmol) of N-iodosuccinimide was solubilized in 150 mL of chloroform/acetic acid (2:1) and added dropwise to the solution during 1 h still at 0 °C. The reaction mixture was allowed to warm up to room temperature during the night (red coloration of the solution). The reaction was poured onto iced water and extracted with hexane. The organics were washed with saturated solution of $Na_2S_2O_3$ and with water and brine. They were dried over Na_2SO_4. The crude product was

purified by column chromatography on silica gel (eluent: hexane). 7.40 g of incolor oil were obtained (yield 88%).

^{1}H NMR ($CDCl_3$): 7.40 (d, 1H, J=5.7 Hz); 6.65 (d, 1H, J=5.7 Hz); 4.00 (t, 2H, J=6.3 Hz); 1.74 (m, 2H); 1.32 (m, 4H) and 0.89 (t, 3H, J=5.1 Hz). ^{13}C NMR ($CDCl_3$): 159.1; 129.7; 116.8; 72.2; 54.6; 31.5; 29.4; 25.5; 22.6 and 14.0.

2.2.6. 2,5 dibromo-3-hexyloxythiophene ***M2***

0.5 g (2.7 mmol) of 3-hexyloxythiophene was diluted in 5 mL of freshly distilled THF under argon at −78 °C. 2.2 equivalents (1.06 g, 5.97 mmol) of N-bromosuccinimide solubilized in 15 mL of anhydrous THF were added dropwise during 30 min to the solution kept at −78 °C. The reaction mixture was allowed to warm up slowly to room temperature over 2 h before being hydrolyzed. The mixture was extracted with hexane. The organics were washed with water and brine. They were dried over Na_2SO_4. The crude product was purified by column chromatography on silica gel using hexane as eluent. 0.82 g of yellow liquid were obtained (yield 88%).

^{1}H NMR ($CDCl_3$): 6.74 (s, 1H); 3.97 (t, 2H, J=7.6 Hz); 1.68 (m, 2H); 1.25 (m, 6H) and 0.87 (t, 3H, J=6.4 Hz). ^{13}C NMR ($CDCl_3$): 153.8; 120.8; 109.6; 90.4; 72.5; 31.5; 29.3; 25.4; 22.7 and 14.0.

2.2.7. 2 bromo-4-hexyloxythiophene ***C5***

5 g (14.6 mmol) of 2,5 dibromo-3-hexyloxythiophene was diluted in 30 mL of anhydrous THF under argon at −78 °C. 1.1 equivalents (6.5 mL of a 2.5 mol L^{-1} solution in hexane, 16.1 mmol) of BuLi were added dropwise to the solution during 30 min still at −78 °C. The reaction mixture was kept at −78 °C for an additional period of 30 min. Once hydrolyzed, the mixture was extracted with hexane. The organics were washed with water and brine. They were dried over Na_2SO_4. The crude product was purified by column chromatography on silica gel using hexane as eluent. A more efficient purification was performed with HPLC (eluent: acetonitrile). 1.0 g of colorless liquid was obtained (yield 26%).

^{1}H NMR ($CDCl_3$): 6.72 (d, 1H, J=2.2 Hz); 6.08 (d, 1H, J=1.9 Hz); 3.86 (t, 2H, J=7.6 Hz); 1.69 (m, 2H); 1.26 (m, 6H) and 0.35 (t, 3H, J=6.4 Hz). ^{13}C NMR ($CDCl_3$): 156.8; 122.6; 111.5; 98.4; 69.97; 31.6; 29.1; 25.7; 22.6 and 14.0.

2.2.8. 3-carbonitrile-3′-hexyl-2,2′-bithiophene ***D1***

2 g (18.3 mmol) of 3-carbonitrilethiophene was diluted in 30 mL of anhydrous diethyl ether. The mixture was cooled down to −78 °C. 1.1 equivalents (20.2 mmol, 13.5 mL of a 1.5 M solution in THF) of LDA were added dropwise. The mixture was allowed to warm up to room temperature during 1 h. Anhydrous $ZnCl_2$ (2.4 eq., 5.8 g, 43.9 mmol) was solubilized in THF (25 mL) under argon and was then added to the mixture at −78 °C. The reaction mixture was kept at −50 °C during 2 h. Still at −50 °C, 1.2 equivalents (5.43 g, 21.96 mmol) of 2-bromo-3-hexylthiophene were added. A solution of $Pd(PPh_3)_3$ in THF was also added (7% molar). The reaction mixture was allowed to warm up to room temperature during 2 h and then heated to reflux during the night. After hydrolysis, the mixture was extracted with 3×100 mL of diethyl ether. The organic layers were collected and dried over Na_2SO_4. After filtration and solvent evaporation, the product was purified by column chromatography on silica gel eluted with hexane/dichloromethane (50/50). A more efficient purification was performed with reverse phase HPLC (eluent: acetonitrile). 1.42 g of yellowish oil was obtained (yield 28%).

^{1}H NMR ($CDCl_3$): 7.37 (d, 1H, J=5.4 Hz); 7.35 (d, 1H, J=5.4 Hz); 7.25 (d, 1H, J=5.4 Hz); 7.00 (d, 1H, J=5.1 Hz); 2.67 (t, 2H, J=7.3 Hz); 1.58 (m, 2H); 1.23 (m, 6H) and 0.84 (t, 3H, J=6.4 Hz). ^{13}C NMR ($CDCl_3$): 146.5; 144.0; 130.0; 129.9; 127.6; 127.2; 125.5; 115.6; 110.1; 32.0; 30.9; 29.6; 29.5; 23.0 and 14.5. Anal. Calcd for $C_{15}H_{17}NS_2$: C 65.41%; H 6.22%; N 5.08% and S 23.28%. Found: C 65.79%; H 6.19%; N 4.82% and S 21.80%.

The same procedure was used to synthesize the monomers **D2–D4**. The yields were 76%, 45.5% and 51.4% for monomers **D2, D3** and **D4,** respectively.

2.2.9. 3-carbonitrile-4′-hexyl-2,2′-bithiophene ***D2***

^{1}H NMR ($CDCl_3$): 7.41 (d, 1H, J=1.1 Hz); 7.17 (s, 2H); 6.98 (d, 1H, J=0.9 Hz); 2.60 (t, 2H, J=7.4 Hz); 1.56 (m, 2H); 1.29 (m, 6H) and 0.87 (t, 3H, J=6.5Hz). ^{1}H NMR (Acetone-d_6): 7.54 (d, 1H, J=5.4 Hz); 7.45 (d, 1H, J=1.4 Hz); 7.29 (d, 1H, J=5.4 Hz); 7.24 (d, 1H, J=1.4 Hz); 2.61 (t, 2H, J=7.5 Hz); 1.61 (m, 2H); 1.27 (m, 6H) and 0.82 (t, 3H, J=6.8 Hz). ^{13}C NMR ($CDCl_3$): 147.3; 144.7; 132.7; 130.0; 128.4; 124.2; 122.4; 115.8; 104.8; 31.6; 30.4; 30.3; 28.9; 22.6 and 14.1. Anal. Calcd for $C_{15}H_{17}NS_2$: C 65.41%; H 6.22%; N 5.08% and S 23.28%. Found: C 65.04%; H 6.09%; N 5.11% and S 22.07%.

2.2.10. 3-carbonitrile-3′-hexyloxy-2,2′-bithiophene ***D3***

^{1}H NMR ($CDCl_3$): 7.30 (d, 1H, J=5.4 Hz); 7.16 (d, 1H, J=5.4 Hz); 7.14 (d, 1H, J=5.4 Hz); 6.88 (d, 1H, J=5.7 Hz); 4.15 (t, 2H, J=6.3 Hz); 1.84 (m, 2H); 1.53–1.32 (m, 4H) and 0.88 (t, 3H, J=6.7 Hz). ^{13}C NMR ($CDCl_3$): 155.8; 144.9; 128.5; 125.2; 123.9; 116.5; 115.9; 111.6; 103.0; 72.1; 31.4; 29.4; 25.6; 22.5 and 14.0. Anal. Calcd for $C_{15}H_{17}NOS_2$: C 61.82%; H 5.88%; N 4.81% and S 22.00%. Found: C 62.00%; H 5.78%; N 4.64% and S 21.82%.

2.2.11. 3-carbonitrile-4′-hexyloxy-2,2′-bithiophene ***D4***

^{1}H NMR ($CDCl_3$): 7.20 (d, 1H, J=1.6 Hz); 7.16 (d, 1H, J=5.4 Hz); 7.15 (d, 1H, J=5.4 Hz); 6.28 (d, 1H, J=1.9 Hz); 3.91 (t, 2H, J=6.3 Hz); 1.73 (m, 2H); 1.32 (m, 4H) and 0.87 (t, 3H, J=6.7 Hz). ^{13}C NMR ($CDCl_3$): 157.7; 146.7; 131.2; 129.9; 124.5; 119.0; 115.3; 104.9; 99.4; 70.17; 31.4; 28.9; 25.5; 22.4 and 13.9. Anal. Calcd for $C_{15}H_{17}NOS_2$: C 61.82%; H 5.88%; N 4.81% and S 22.00%. Found: C 57.46%; H 5.24%; N 4.17% and S 21.69%.

2.3. Polymer syntheses

The general procedure of polymerization was as follows. A solution of monomer (1 eq. of **D1**–**D4)** in anhydrous chloroform (3 mL) was added dropwise to a solution of anhydrous ferric chloride (4 eq., 2.91 mmol, 471 mg) in anhydrous chloroform (6 mL). The addition was performed at 0 °C with constant stirring over a period of 30 min. At the end of the addition, the mixture was allowed to warm up to room temperature for 24 h. The mixture was then poured onto methanol. The crude polymer was thus filtered and washed thoroughly with aqueous hydrazine and washed with an aqueous solution of ammonia (0.1 mol L^{-1}) and of ethylenediaminetetraacetic acid (0.05 mol L^{-1}) successively. The material was then filtered and washed several times with water and methanol. The polymer mixture was then fractionated in a Soxhlet apparatus using the following sequence of solvents: methanol, hexane, dichloromethane, chloroform and toluene if necessary. For each fraction, the solvent was evaporated and then the polymer collected was analysed by SEC to estimate its weight-average (Mw) molecular weight (see supporting informations Table I). For all polymers, the dichloromethane extract constituted the one abundant fraction and these fractions were considered as the purified polymer. The overall yield of the polymerization reaction was 89%, 63%, 84% and 27% for $\mathbf{P_{D1}}$, $\mathbf{P_{D2}}$, $\mathbf{P_{D3}}$ and $\mathbf{P_{D4}}$, respectively. ^{1}H-RMN spectra were recorded from CH_2Cl_2 fraction.

Table 1
UV-visible characteristics for absorption spectrum of the P3HT, and $\mathbf{P_{D1}}$–$\mathbf{P_{D4}}$ in chlorobenzene solution and as film obtained by spin coating from chlorobenzene solution.

Polymer		Absorption in solution λ_{max} (nm)	Absorption as film λ_{max} (nm) (relative intensity)		
$\mathbf{P_{D1}}$	CH_2Cl_2	425	500 (1.00)	549 (0.84)	590 (0.58)
	$CHCl_3$	431	516 (1.00)	552 (0.99)	601 (0.78)
$\mathbf{P_{D2}}$	CH_2Cl_2	444	489 (1.00)	548 (0.83)	608 (0.46)
	$CHCl_3$	478	519 (0.94)	556 (1.00)	602 (0.87)
$\mathbf{P_{D3}}$	CH_2Cl_2	503	510 (1.00)	552 (0.84)	-
	$CHCl_3$	516	536 (1.00)	573 (0.92)	639 (0.58)
$\mathbf{P_{D4}}$	CH_2Cl_2	540	554 (1.00)	–	–
	$CHCl_3$	548	–	–	–
P3HT	batch	456	528 (0.98)	552 (1.00)	602 (0.69)

2.3.1. Poly(3-carbonitrile-3′-hexyl-2,2′-bithiophene) – $\mathbf{P_{D1}}$

^{1}H NMR (70 °C, $C_2D_2Cl_4$): 7.75 (s, 0.3H), 7.58 (s, 0.3H), 7.4 (s, 1H), 7.2 (s, 1H), 2.78 (m, 2H), 1.71–0.91 (m, 11H).

SEC analysis (CH_2Cl_2 fraction): M_n=2820 g mol^{-1}, M_w=3480 g mol^{-1} and IP=1.5

2.3.2. Poly(3-carbonitrile-4′-hexyl-2,2′-bithiophene) – $\mathbf{P_{D2}}$

^{1}H NMR (70 °C, $C_2D_2Cl_4$): 7.58 (s, 0.2H), 7.53 (s, 1H), 7.48 (s, 0.1H), 7.32 (s, 1H), 2.84 (t, 2H), and 1.75–0.96 (m, 11H).

SEC analysis (CH_2Cl_2 fraction): M_n=2060 g mol^{-1}, M_w=2360 g mol^{-1} and IP=1.1

2.3.3. Poly(3-carbonitrile-3′-hexyloxy-2,2′-bithiophene)—$\mathbf{P_{D3}}$

^{1}H NMR (70 °C, $C_2D_2Cl_4$): 6.95–7.41 (m, 2.5H), 4.27 (t, 2H), 1.95 (m, 2H), 1.44 (m, 8H) and 0.97 (m, 3H).

SEC analysis (CH_2Cl_2 fraction): M_n=1350 g mol^{-1}, M_w=1490 g mol^{-1} and IP=1.1

2.3.4. Poly(3-carbonitrile-4′-hexyloxy-2,2′-bithiophene) – $\mathbf{P_{D4}}$

^{1}H NMR (70 °C, $C_2D_2Cl_4$): 7.33–7.37 (m, 2H), 4.27 (m, 2H), 1.97 (m, 2H), and 0.96–1.32 (m, 7H).

SEC analysis (CH_2Cl_2 fraction): M_n=1270 g mol^{-1}, M_w=1720 g mol^{-1} and IP=1.3

3. Results and discussion

3.1. Synthesis

The synthesis routes to bithiophene monomers are shown in Schemes 1 and 2 according to an approach described by Demanze et al. [21]. Four monomers were prepared with head-to-head or head-to-tail coupling between a carbonitrilethiophene and an alkyl or alkoxythiophene. The key step for the synthesis of unsymmetrically disubstituted bithiophene **D1–D4** involves the coupling of an organozincic derivative, the 2-chlorozinc-3-carbonitrilethiophene with a 2-halogenated or 5-halogenated thiophene derivative. This coupling is realised through the Negishi reaction using a Pd(0) complex as catalyst. As expected, the yield of the palladium-catalysed coupling reaction is improved when the halogenated thiophene is an iodo derivative. For the monomer **D1**, the coupling of 2-chlorozinc-3-carbonitrilethiophene with 2-bromo-3-hexylthiophene **C1** led to a yield of 28%, which increased to 39% when the 2-iodo-3-hexylthiophene **C2** is used as the halogenated derivative. In addition, the comparison of the yields achieved for monomers **D1**–**D2** and **D3**–**D4** shows that the coupling reaction yield is higher for the head-to-tail bithiophene than for the head-to-head dimer. This is attributed to the decrease in the steric hindrance in the head-to-tail coupling in comparison with the head-to-head one.

From these four monomers, alternate copolymers $\mathbf{P_{D1}}$–$\mathbf{P_{D4}}$ were prepared by chemical oxidation of the corresponding monomers in presence of $FeCl_3$ according to a method similar to the one used by Sugimoto et al. [22] This procedure does not allow controlling the regioregularity of the coupling between repeating units. The polymer can be considered as regiorandom. After the polymerization, the crude polymer was purified and fractioned by Soxhlet extraction with methanol, hexane, dichloromethane and chloroform. The molecular weigths of each polymer extracted of these different fractions appear relatively low (see Supporting Informations Table I). Several hypotheses could be given. On the first hand, these copolymers own only one alkyl lateral chain for two thiophene units, which limit their solubility and thus the growth of the polymer chain is induced by their precipitation. On the second hand, the presence of the carbonitrile groups and their strong electron-withdrawing behaviour could deactivate the α-position and limit the polymerization process leading to low molecular weight polymers [23,24]. Most of the following characterizations have been carried out on dichloromethane extracts since they constituted the most abundant fractions.

3.2. Adsorption spectra

The electronic absorption data of the various polymers are listed in Table 1. All spectroscopic properties were measured in dilute chlorobenzene solution as well as in solid thin film at room temperature.

The UV–vis adsorption spectra of various polymers in solution are shown in Fig. 1A. The adsorption maximum (λ_{max}) of alkylthiophene polymer $\mathbf{P_{D2}}$ (444 nm) is red-shifted in comparison with the one of polymer $\mathbf{P_{D1}}$ (425 nm). This should be attributed to the difference in regioregularity of the polymers. It is well-known that HT regioregular polythiophenes have a more planar conformation than HH regioregular polythiophenes. This effect is also observed for the polymers containing alkoxy chains. The adsorption maximum of $\mathbf{P_{D4}}$ is centred at 540 nm, which is 37 nm red-shifted compared to the λ_{max} (503 nm) of polymer $\mathbf{P_{D3}}$. Furthermore the comparison of the absorption spectra of polymers $\mathbf{P_{D1}}$ and $\mathbf{P_{D3}}$ shows that the maximum absorption of $\mathbf{P_{D3}}$ exhibits a 78 nm red shift as compared to $\mathbf{P_{D1}}$. This absorbance shift is even more significant for the polymers $\mathbf{P_{D4}}$ and $\mathbf{P_{D2}}$ since their λ_{max} is at 540 and 444 nm, respectively. The adsorption behaviour of $\mathbf{P_{D3}}$ and $\mathbf{P_{D4}}$ appears as a result of the electro-donating effect created by the introduction of hexyloxy side chains. In combining a regioregular structure by head-to-tail couplings and the presence of the electron-donor moiety, the absorbance of the polymers in solution is substantially red-shifted as illustrated by the shift of the λ_{max} values from $\mathbf{P_{D1}}$ (425 nm) to $\mathbf{P_{D4}}$ (540 nm) in spite of the presence of electron-withdrawing carbonitrile groups on thiophene units.

This absorbance shift is less visible in the solid state for dichloromethane soluble fractions (Fig. 1B, Table 1). Indeed the absorption maxima of the polymers $\mathbf{P_{D1}}$ and $\mathbf{P_{D4}}$ as thin films are

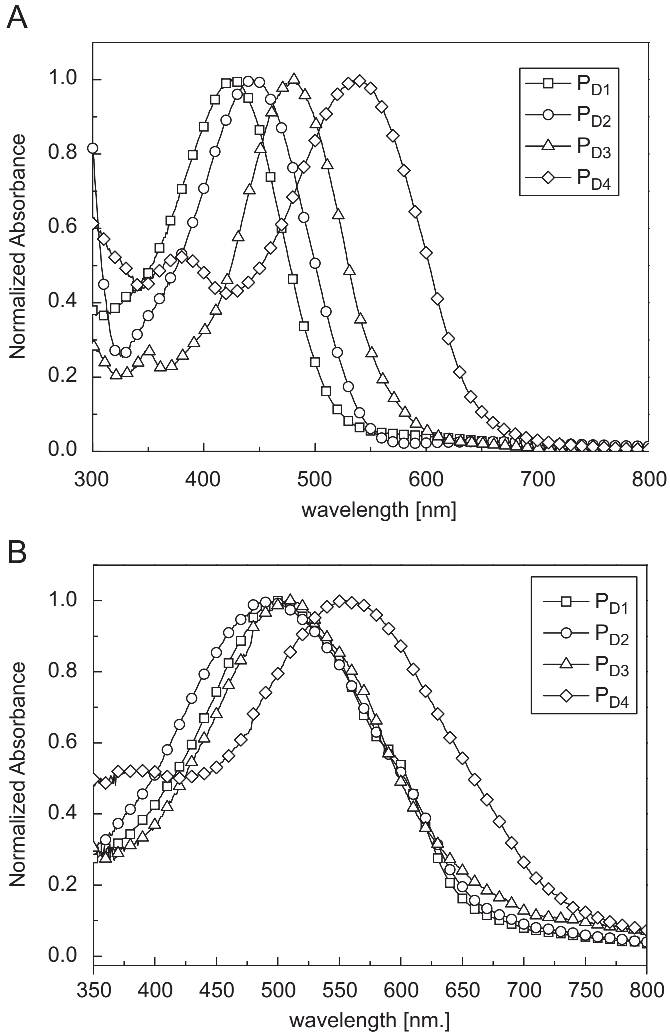


Fig. 1. Normalized UV–vis absorption spectra of the dichloromethane fraction of $\mathbf{P_{D1}}$–$\mathbf{P_{D4}}$ in solution (A) and in the solid state (B) obtained by spin coating from chlorobenzene solution.

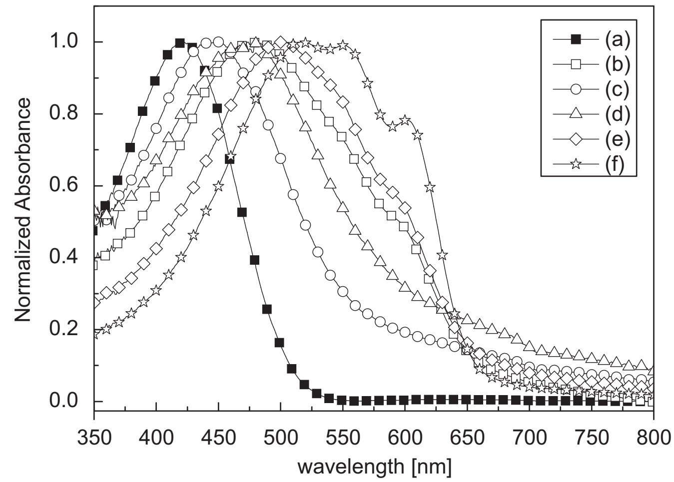


Fig. 2. Normalized UV–vis absorption spectra of $\mathbf{P_{D1}}$ in chlorobenzene solution (a) the batch fraction, and in the solid state obtained by spin coating from chlorobenzene solution from (b) the batch fraction, (c) the MeOH fraction, (d) the hexane fraction, (e) the dichloromethane fraction and (f) the chloroform fraction.

around 500 and 554 nm, respectively. Nevertheless for all polymers, the maximum peak in the solid state is red-shifted in comparison with the one in solution. This effect is more significant for the polymers $\mathbf{P_{D1}}$ and $\mathbf{P_{D2}}$ by about 75 and 45 nm, respectively than for the polymers $\mathbf{P_{D3}}$ and $\mathbf{P_{D4}}$ (7 and 10 nm). The bathochromic shift upon passing from solution to the solid state is presumably due to the increased π–π stacking interactions in the solid state, a phenomenon commonly observed in conjugated polymers. When comparing the spectrum of $\mathbf{P_{D4}}$ as thin film to those of other polymers, $\mathbf{P_{D4}}$ shows a broader absorption band in the long-wavelength region. This red-shift and band-broadening effect could be ascribed to the more extended conjugation and planarity of polymer chains allowing an efficient packing. As shown in Fig. 2 in the case of polymer $\mathbf{P_{D1}}$, the appearance of the vibronic structures on the absorption spectrum is even more visible for a film elaborated from the choroform fraction containing the polymer of higher molecular weight. The vibronic structures (516, 552, and 601 nm) are observed in both cases for dichloromethane and chloroform soluble fractions. The lower molecular weight fraction (dichloromethane fraction shown in Fig. 1B) did not get organized as well as the higher ones. This difference in structuration was also observed on the AFM images (see Supporting Informations Fig. D) where fibrillar structures were obtained from chloroform fraction and not for dichloromethane fraction.

3.3. Electrochemical properties

Electrochemical cyclic voltammetry is often used to measure the redox properties of the conjugated polymers from which the level of the highest occupied molecular orbital (HOMO) and the lowest unoccupied molecular orbital (LUMO) of the polymers are estimated. For each polymer $\mathbf{P_{D1}}$–$\mathbf{P_{D4}}$, the cyclic voltammogram (CV) was recorded in an anhydrous propylene carbonate (PC) solution containing 0.1 mol L^{-1} of $TBABF_4$ salt. The CVs are obtained from a film deposited on a gold (Au) electrode by drop casting of a solution of the polymer solubilised in chloroform (Fig. 3). The CV data of all the polymers are gathered in Table 2.

All polymers present an onset reduction potential (E^{onset}_{red}) around -1.60 V versus Ag/Ag^+ except for the polymer $\mathbf{P_{D3}}$ whose reduction takes place at a more negative potential (-1.83 V). However, the polymers $\mathbf{P_{D1}}$–$\mathbf{P_{D4}}$ were reduced in a potential range much more anodic in comparison with the onset reduction potential value obtained for the P3HT in the same conditions (-2.55 V). This effect could be assigned to the presence of carbonitrile groups, which decrease the electronic density of the π conjugated system leading to an anodic shift of the reduction potential compared to the P3HT.

The effects of the hexyloxy side chains and of the regioregularity (HH or HT coupling) on the electrochemical behaviour of $\mathbf{P_{D1}}$–$\mathbf{P_{D4}}$ were observed through the onset oxidation potential values (E^{onset}_{ox}) of these polymers, which are estimated from CVs shown in Fig. 3. Indeed, the oxidation onsets of the polymers $\mathbf{P_{D3}}$ and $\mathbf{P_{D4}}$ take place at a potential much more cathodic than for $\mathbf{P_{D1}}$ and $\mathbf{P_{D2}}$, i.e. 0.35, 0.18 and 0.61, 0.57 V respectively (Table 2). This could be explained by the electron-donor effect of the hexyloxy chains leading to the lower oxidation potentials E^{onset}_{ox} of $\mathbf{P_{D3}}$ and $\mathbf{P_{D4}}$ compared to the ones of $\mathbf{P_{D1}}$ and $\mathbf{P_{D2}}$. Concerning the role of the regioregularity, one can observe that the E^{onset}_{ox} value of $\mathbf{P_{D4}}$ was cathodically shifted to0.17 V compared to the one of $\mathbf{P_{D3}}$. For $\mathbf{P_{D4}}$, the HT coupling led to a more delocalized π conjugated backbone than for $\mathbf{P_{D3}}$ where the HH coupling present in every repeating units induced steric hindrances between adjacent hexyloxy chains that limited the delocalized π conjugated backbone. The greater is the degree of regioregularity,

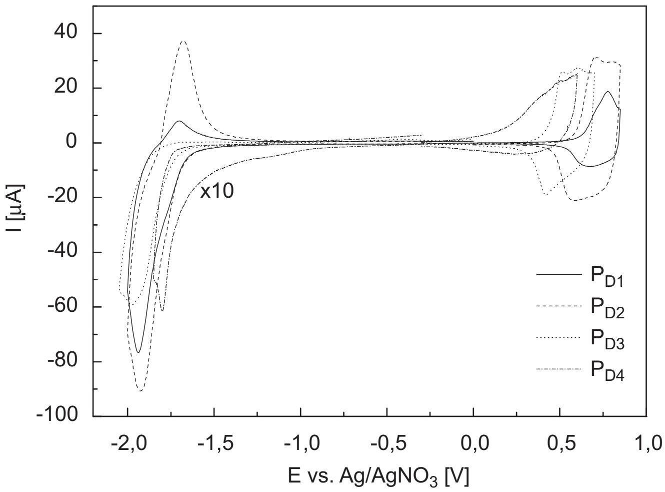


Fig. 3. Cyclic voltammograms in propylene carbonate (PC+TBABF$_4$ 0.1 M) at 5 mV s^{-1} of the films $\mathbf{P_{D1}}$–$\mathbf{P_{D4}}$ obtained by drop casting onto Au electrode (S=7 × 10^{-2} cm^2) from a chloroform solution of polymer.

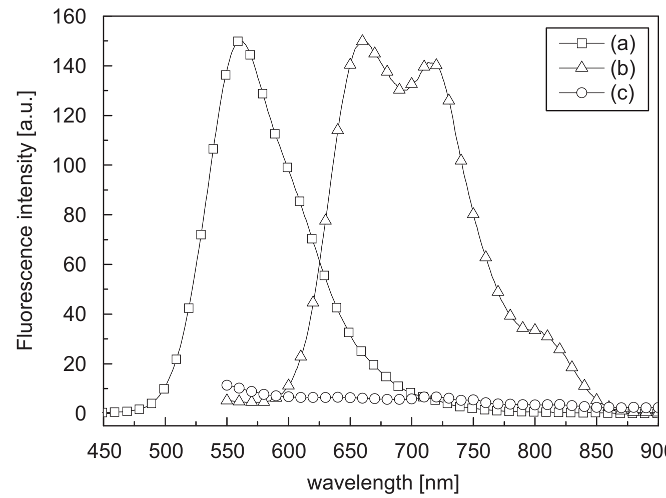


Fig. 4. Fluorescence emission spectra for the $\mathbf{P_{D1}}$ (a) in chlorobenzene solution (b) as thin film and (c) as thin film in blend with PCBM with 1:1 weight ratio.

Table 2
Electrochemical data of the polymers obtained chemically.

Polymer	E_{ox}^{onset}(V)	E_{red}^{onset} (V)	HOMO (eV)	LUMO (eV)	$\Delta E^{electroch}$ (eV)	ΔE^{opt} (eV)
$\mathbf{P_{D1}}$	0.61	−1.65	−5.35	−3.08	2.26	2.00
$\mathbf{P_{D2}}$	0.57	−1.65	−5.34	−3.08	2.22	1.95
$\mathbf{P_{D3}}$	0.35	−1.83	−5.15	−2.96	2.18	1.82
$\mathbf{P_{D4}}$	0.18	−1.63	−4.91	−3.10	1.81	1.80

the longer is the conjugation length ,which leads to the decrease in the oxidation potential. This effect is even more visible in the case of polymers $\mathbf{P_{D1}}$ and $\mathbf{P_{D2}}$.

The HOMO and the LUMO energy levels and the electrochemical band gaps ($E_g^{EC}=|E_{ox}^{onset}-E_{red}^{onset}/eV$) of polymers $\mathbf{P_{D1}}$–$\mathbf{P_{D4}}$ were calculated from the E_{ox}^{onset} and E_{red}^{onset} values listed in Table 2. It can been seen that the LUMO energy level are almost the same for all polymers around −3.10 eV while the HOMO energy levels vary between −5.35 and −4.91 eV from $\mathbf{P_{D1}}$ to $\mathbf{P_{D4}}$. As a consequence, the electrochemical bandgaps decrease from 2.26 eV to 1.81 eV and are lower than the one of P3HT (2.80 eV) for every polymer. Overall, as we were expecting, all four polymers show low band gaps. By increasing the electron-donor effect of chains and the control of the regioregularity, the electrochemical band gap decrease from $\mathbf{P_{D1}} > \mathbf{P_{D2}} > \mathbf{P_{D3}} > \mathbf{P_{D4}}$. This is in agreement with the red-shift of adsorption spectra shown in Fig. 1a and with the optical band gaps (Eg^{opt}) calculated from the adsorption and fluorescence spectra in the solid state (Table 2). For all polymers, it appears that the electrochemical band gaps were larger than those of the optical band gaps as usually observed for other polythiophene derivatives.

3.4. Photovoltaic characterization

Before testing the $\mathbf{P_{D1}}$–$\mathbf{P_{D4}}$ polymers in blends with PCBM in bulk heterojunction photovoltaic devices, fluorescence measurements (PL) were performed to examine the charge transfer efficiency from the polymer to the PCBM. $\mathbf{P_{D1}}$ was used as a model of these measurements. As shown in Fig. 4, there is a drastic decrease in the emission fluorescence of $\mathbf{P_{D1}}$ in the solid state by addition of the PCBM for a 1:1 weight ratio and also for various compositions of $\mathbf{P_{D1}}$/PCBM (1:0.5 and 1:1.5). The significant fluorescence quenching is a strong indication that in the $\mathbf{P_{D1}}$:PCBM system, the charge transfer from the photoexcited conjugated backbone to PCBM is efficient and thus this system may be used in photovoltaic applications. However, one notes that the absorption spectrum (Fig. 5A) in solid state of $\mathbf{P_{D1}}$:PCBM blend films are different by comparison with the one of $\mathbf{P_{D1}}$. Besides an absorption band at 350 nm due to PCBM, it appears that the absorption maximum of $\mathbf{P_{D1}}$ (λ_{max}) was blue shifted and that the intensity of bands at 541 and 592 nm decrease with larger amounts of PCBM. The higher the quantity of PCBM, the lower is λ_{max} and the intensity of the vibronic structures. This suggests that the PCBM limits the good organisation of polymer chains [25]. On the other hand, the solid state absorption spectrum of $\mathbf{P_{D3}}$ in a blend with PCBM remains unchanged for ratio in PCBM up to 1.2 (Fig. 5B). The molecular organisation of $\mathbf{P_{D3}}$ film was kept unchanged in the mixture. The presence of alkoxy chains led to more structured material through the S–O interaction, which permits to obtain more planar polymer chains [10,26] and thus more structured materials even in the presence of PCBM. Indeed preliminary diffraction X-measurements on powder and film performed on $\mathbf{P_{D1}}$ and $\mathbf{P_{D3}}$ show that the alkoxy-containing material is much more structured in comparison with the alkyl-containing one (see Supporting Informations Figure A).

To check the photovoltaic properties of $\mathbf{P_{D1}}$–$\mathbf{P_{D4}}$ polymers, we fabricated polymer solar cells with the architecture glass-ITO/PEDOT:PSS/polymer:PCBM blend/LiF/Al where the polymer was used as the electron donor and PCBM as the electron acceptor. The active layer constituted of a polymer:PCBM blend at various polymer:PCBM weight ratios and was deposited by spin coating on an ITO/glass substrate covered by a PEDOT:PSS thin layer.

The current–voltage characteristics of the solar cells under illumination of AM 1.5, (100 mW cm^{-2}) are shown in Fig. 6. Table 3 lists the photovoltaic properties obtained from the J–V curves for the best devices. The optimization of the active layer in terms of the polymer concentration in chlorobenzene solution and the polymer:PCBM ratio used to make the polymer:PCBM blend has been carried out in order to achieve the best devices. For each polymer the optimal concentration was found to be 10 g L^{-1} (see Supporting Informations Fig. B) and the optimum polymer:PCBM ratio was around 1:1 in weight.

Devices constructed from both polymers clearly show a rectifying behaviour and exhibit significant photovoltaic effects

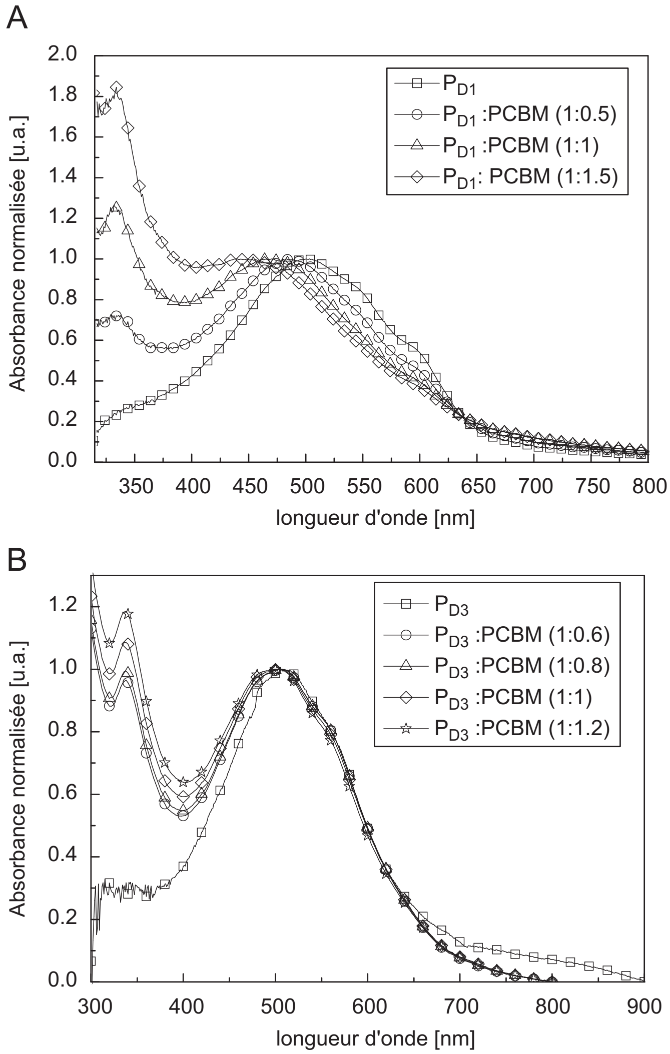


Fig. 5. Evolution of absorption spectra of (A) $\mathbf{P_{D1}}$ (CH_2Cl_2 fraction) and (B) $\mathbf{P_{D3}}$ (CH_2Cl_2 fraction), as a film obtained from a chlorobenzene solution (10 g L^{-1}), alone and in blend with PCBM at different ratios.

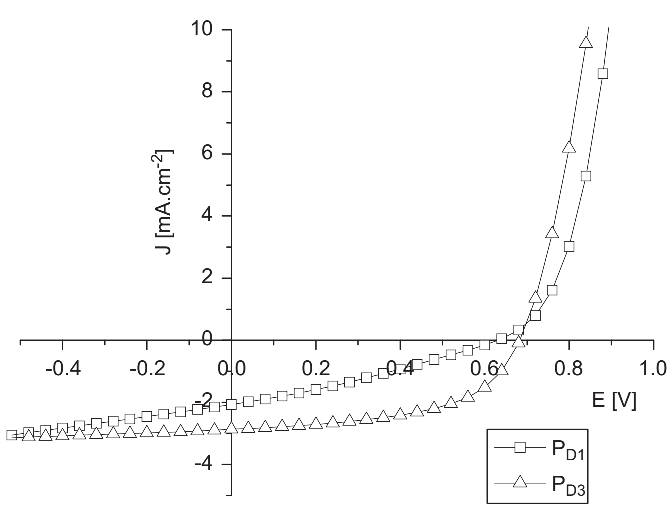


Fig. 6. Current density–voltage curves for pristine active layer (AM 1.5, 100 mW cm^{-2}) of $\mathbf{P_{D1}}$ and $\mathbf{P_{D3}}$.

under illumination. Our measurements (Table 3) indicate that these devices have a high open-circuit voltage in comparison with the one obtained from P3HT:PCBM cells (Voc ~ 0.55–0.60 V). The decrease in the V_{oc} from $\mathbf{P_{D1}}$ to $\mathbf{P_{D4}}$ is in agreement with the value of the HOMO levels of these polymers (Table 2) since the V_{oc} is related to the energy difference between the LUMO of acceptor (PCBM) and the HOMO of the donor (conjugated polymer) [8]. It can also been seen that the current at the short-circuit (J_{SC}) and the power conversion efficiency (PCE) are increasing from $\mathbf{P_{D1}}$ to $\mathbf{P_{D3}}$. The J_{SC} and PCE of the device based on $\mathbf{P_{D3}}$ reached 2.87 mA cm^{-2} and 1.07% respectively, which are around twice time of those of the cells based on $\mathbf{P_{D1}}$ (1.81 mA cm^{-2} and 0.52%). Furthermore, the devices based on $\mathbf{P_{D3}}$ presents a better FF of 0.543. The improvement in J_{SC} and PCE could be ascribed to a better charge separation and transport between $\mathbf{P_{D3}}$ and PCBM. By comparison with $\mathbf{P_{D1}}$, this result should be attributed to the well-organized structure of $\mathbf{P_{D3}}$ conserved in the $\mathbf{P_{D3}}$:PCBM blend film contrary to $\mathbf{P_{D1}}$. Thanks to the S–O interaction, which permits to maintain the more planar polymer chains. Interestingly, the performance parameters of the device based on $\mathbf{P_{D2}}$ were just between those of devices based on $\mathbf{P_{D1}}$ and $\mathbf{P_{D3}}$. Indeed the power conversion efficiency for the devices based on $\mathbf{P_{D2}}$ reached 0.70%, which is nearly 1.5 times higher than one of the device based on $\mathbf{P_{D1}}$. As previously described, the better performances are obtained with the polymer showing better regioregular structure, i.e. the head-to-tail coupled polymer $\mathbf{P_{D2}}$. It is noteworthy that the cells based on $\mathbf{P_{D2}}$ and $\mathbf{P_{D3}}$ show a similar short-circuit current density (2.84 and 2.87 mA cm^{-2}, respectively) but the device make of the bulk heterojunction composite of $\mathbf{P_{D3}}$ exhibits much higher fill factor than the cell prepared with polymer $\mathbf{P_{D2}}$. This could be the result of a more efficient charge extraction for the cell based on $\mathbf{P_{D3}}$ in comparison with the $\mathbf{P_{D2}}$ device.

Table 3
Best performances obtained with each polymer (dichloromethane fraction) in photovoltaic devices.

	$\mathbf{P_{D1}}$	$\mathbf{P_{D2}}$	$\mathbf{P_{D3}}$	$\mathbf{P_{D4}}$
Concentration (g L^{-1})	10	10	10	10
Polymer:PCBM ratio	1:0.8	1:1	1:1.2	1:1
Thickness (nm)	45	45	50	45
Voc (V)	0.797	0.694	0.683	0.518
Jsc (mA cm^{-2})	1.81	2.84	2.87	1.02
FF	0.362	0.357	0.543	0.282
PCE (%)	**0.52**	**0.70**	**1.07**	**0.15**

However, we observe a significant decrease in the PCE for the cells based on $\mathbf{P_{D4}}$. Indeed the J_{SC}, V_{OC} and FF values are obviously lower than those of other devices. These unexpected results might be ascribed to the poor charge separation (and/or recombination) and transportation and also influenced by the work function of electrodes of the device as well as the film morphology. The HOMO level of P_{D4} is however above the work function of PEDOT:PSS (−5.1 eV) and ITO. The poor solar cell performances are disappointing in spite of both the lower band gap of $\mathbf{P_{D4}}$ and the regioregular and planar structure of the polymer backbone. This is not well-understood at present. It is noteworthy that the absorption spectra of $\mathbf{P_{D4}}$:PCBM blend in the solid state remains practically identical to the one measured for $\mathbf{P_{D4}}$ as film (see Supporting Informations Figure C). In general, for all type of polymers, the thicknesses of the best devices are quite low around 50 nm. This means certainly that the mobilities of the materials are too low. Some transistor or ToF measurements could be done to explain these results and perhaps also explain the differences observed for $\mathbf{P_{D4}}$.

Moreover, optimization of the device parameters by thermal annealing did not show an increase in the final photovoltaic performance for the device using $\mathbf{P_{D4}}$ as donor. The same result has been also achieved in the case of polymers $\mathbf{P_{D1}}$–$\mathbf{P_{D3}}$.

4. Conclusion

Four polymers containing alternating electron-donating (hexyl ou hexyloxy) and electron-withdrawing (carbonitrile) moieties have been synthesized by $FeCl_3$ oxidative polymerization from disubstituted bithiophene monomers (**D1**–**D4**). By cyclic voltammetry, the influence of carbonitrile groups on polythiophene chains was clearly shown since the LUMO level of polymers $\mathbf{P_{D1}}$–$\mathbf{P_{D4}}$ is strongly decreased at around -3.10 eV in comparison with the one of the P3HT (-2.19 eV). Furthermore, the HOMO level of the polymer is increased from $\mathbf{P_{D1}}$ to $\mathbf{P_{D4}}$. This could be seen as a result of both the effect induced by the electron-donating (hexyl or hexyloxy) chains and the planarity of the polythiophene chains controlled through the regioregularity of bithiophene monomers—precursors of the polymers. The polymer prepared from bithiophenes synthesized by a head-to-tail (HT) coupling shows a HOMO level slightly higher than one of the polymer elaborated from head-to-head (HH) bithiophenes. Nevertheless, the effect of the planarity appears more clearly when the hexyl units linked to polythiophene backbone ($\mathbf{P_{D1}}$ and $\mathbf{P_{D2}}$) are substituted by hexyloxy moieties ($\mathbf{P_{D3}}$ and $\mathbf{P_{D4}}$). The presence of alkoxyl substituents leads to an increase in the HOMO level from ~ -5.3 eV for $\mathbf{P_{D1}}$ and $\mathbf{P_{D2}}$ to ~ -5.0 eV for $\mathbf{P_{D3}}$ and $\mathbf{P_{D4}}$. The reason of this variation is related to a higher planarization of the polythiophene backbone in $\mathbf{P_{D3}}$ and $\mathbf{P_{D4}}$ thanks to S–O interactions, which hold two adjacent thiophene moieties in a more planar conformation. Also, we were able to show that the introduction of carbonitrile groups associated with the control of the polymer planarity allows the modulation of the HOMO and the LUMO levels to give low band gap polymers confirmed by their optical properties.

Bulk heterojunction solar cells were fabricated using blends of polymers $\mathbf{P_{D1}}$–$\mathbf{P_{D4}}$ and PCBM. The organic solar cells tested on polymers $\mathbf{P_{D1}}$–$\mathbf{P_{D3}}$ give power conversion efficiencies up to 1.07%. The best performances were obtained with $\mathbf{P_{D3}}$ containing hexyloxy lateral group and having a lower electrochemical band gap (2.18 eV) in comparison with the one of $\mathbf{P_{D1}}$ and $\mathbf{P_{D2}}$ (2.26 and 2.22 eV, respectively). As unexpected, solar cells made of the polymer $\mathbf{P_{D4}}$ showed low efficiencies although this polymer possess desirable HOMO/LUMO levels with an electrochemical band gap of 1.81 eV. This is not well-understood at present and requires more experiments to give a reasonable explanation. However, the results achieved with $\mathbf{P_{D3}}$ indicate that this type of polymer could be interesting as photovoltaic materials. Although the PCE are quite low for the moment in comparison with the 5.1% obtained in the laboratory with the P3HT:PCBM active layer, some improvements can be done in synthesis method in order to obtain higher molecular weight and more regioregular polymers.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.solmat.2009.12.028.

References

[1] V. Coropceanu, J. Cornil, D.A. daSilvaFilho, Y. Olivier, R. Silbey, J.L. Bredas, Charge transport in organic semiconductors, Chem. Rev. 107 (2007) 926–952.
[2] A.R. Murphy, J.M.J. Frechet, Organic semiconducting oligomers for use in thin film transistors, Chem. Rev. 107 (2007) 1066–1096.
[3] E. Bundgaard, F.C. Krebs, Low band gap polymers for organic photovoltaics, Sol. Energy Mater. Sol. Cells 91 (2007) 954–985.
[4] S. Günes, H. Neugebauer, N.S. Sariciftci, Conjugated polymer-based organic solar cells, Chem. Rev. 107 (2007) 1324–1338.
[5] G. Li, V. Shrotriya, J. Huang, Y. Yao, T. Moriarty, K. Emery, Y. Yang, High efficiency solution processable polymer photovoltaic cells by self-organization of polymer blends, Nat. Mater. 4 (2005) 864–868.
[6] W. Ma, C. Yang, X. Gong, K. Lee, A.J. Heeger, Thermally stable, efficient polymer solar cells with nanoscale control of interpenetrating network morphology, Adv. Funct. Mater. 15 (2005) 1617–1622.
[7] M. Reyes-Reyes, K. Kim, J. Dewald, R. Lopez-Sandoval, A. Avadhanula, S. Curran, D.L. Caroll, Meso-structure formation for enhanced organic cells, Org. Lett. 7 (2005) 5749–5752.
[8] M.C. Scharber, D. Mühlbacher, M. Koppe, P. Denk, C. Waldauf, A.J. Heeger, C.J. Brabec, Design rules for donors in bulk heterojunction solar cells—towards 10% energy-conversion efficiency, Adv. Mater. 18 (2006) 789–794.
[9] C. Shi, Y. Yao, Y. Yang, Q. Pei, Regioregular copolymers of 3-alkoxythiophene and their photovoltaic application, J. Am. Chem. Soc. 128 (2006) 8980–8986.
[10] J. Roncali, P. Blanchard, P. Frère, 3,4-Ethylenedioxythiophene (EDOT) as a versatile building block for advanced functional pi-conjugated systems, J. Mater. Chem. 15 (2005) 1589–1610.
[11] K. Colladet, S. Fourier, T.J. Cleij, L. Lutsen, J. Gelan, D. Vanderzande, L.H. Nguyen, H. Neugebauer, S. Sariciftci, A. Aguirre, G. Janssen, E. Goovaerts, Low band gap donor-acceptor conjugated polymers toward organic solar cells applications, Macromolecules 40 (2007) 65–72.
[12] Y. Li, Y. Zou, Conjugated polymer photovoltaic materials with broad absorption band and high charge carrier mobility, Adv. Mater. 20 (2008) 2952–2958.
[13] F.C. Krebs, Bundgaard, Low band gap conjugated polymers based on thiophene, benzothiadiazole, and benzobis(thiadiazole), Macromolecules 39 (2006) 2823–2831.
[14] M. Chen, E. Perzon, M.R. Andersson, S. Marcinkevicius, S.K.M. Jönsson, M. Fahlman, M. Berggren, 1 micron wavelength photo- and electroluminescence from a conjugated polymer, Appl. Phys. Lett. 84 (2004) 3570–3572.
[15] I. McCulloch, M. Heeney, C. Bailey, K. Genevicius, I. MacDonald, M. Shkunov, D. Sparrowe, S. Tierney, R. Wagner, W. Zhang, M.L. Chabinyc, R.J. Kline, M.D. McGehee, M.F. Toney, Liquid crystalline semiconducting polymers with high charge carrier mobility, Nat. Mater. 5 (2006) 328–333.
[16] B.S. Ong, Y. Wu, P. Liu, S. Gardner, High performance semiconducting polythiophenes for organic thin film transistor, J. Am. Chem. Soc. 126 (2004) 3378–3379.
[17] E.M. Galand, Y.-G. Kim, J.K. Mwaura, A.G. Jones, T.D. McCarley, V. Shrotriya, Y. Yang, J.R. Reynolds, Optimization of narrow band-gap propylenedioxythiophene:cyanovinylene copolymers for optoelectronic application, Macromolecules 39 (2006) 9132–9142.
[18] P. Taranekar, M. Abdulbaki, R. Krishnamoorti, S. Phanichphant, P. Waenkaew, D. Patton, T. Fulghum, R. Advincula, Structure and bandgap design of a new series of light-emitting poly(cyanofluorene-alt-o/m/p-phenylenevinylene)-based copolymers for light-emitting diodes, Macromolecules 39 (2006) 3848–3854.
[19] Y. Zou, J. Hou, C. Yang, Y. Li, A novel n-type conjugated polymer DOCN-PPV:synthesis, optical, and electrochemical properties, Macromolecules 39 (2006) 8889–8891.
[20] J. Cornil, D.A. dos Santos, D. Beljonne, J.-L. Brédas, Electronic structure of phenylene vinylene oligomers : influence of donor/acceptor substitutions, J. Phys. Chem. 99 (1995) 5604–5611.
[21] F. Demanze, A. Yassar, F. Garnier, Alternating donor-acceptor substitutions in conjugated polythiophenes, Macromolecules 29 (1996) 4267–4273.
[22] R. Sugimoto, S. Takeda, H.B. Gu, K. Yoshino, Preparation of soluble polythiophene derivatives utilizing transition metal halides as catalysts and their property, Chem. Express 1 (1986) 635–638.
[23] F. Demanze, P. Godillot, F. Garnier, P. Hapiot, Electrochemical oxidative coupling of cyano-substituted oligothiophenes, J. Electroanal. Chem. 414 (1996) 61–67.
[24] P. Hapiot, F. Demanze, A. Yassar, F. Garnier, Molecular engineering of band level energies in oligothiophenes, through cyano-substitutions, J. Phys. Chem. 100 (1996) 8397–8401.
[25] T. Erb, U. Zhokhavets, G. Gobsch, S. Raleva, B. Stuhn, P. Schilinsky, C. Waldauf, C.J. Brabec, Correlation between structural and optical properties of composite polymer/fullerene films for organic solar cells, Adv. Funct. Mater. 15 (2005) 1193–1196.
[26] J.-M. Raimundo, P. Blanchard, P. Frère, N. Mercier, I. Ledoux-Rak, R. Hierle, J. Roncali, Push–pull chromophores based on 2,2'-bi(3,4-ethylenedioxythiophene) (BEDOT) pi-conjugating spacer, Tetrahedron Lett. 42 (2001) 1507–1510.